

SIMATS ENGINEERING

CO4-AT3

COURSE NAME: Ethical Hacking COURSE CODE: ITA1407

1. Windows Domain Network — Attack Spread

Total systems = 150

Initial exploitation

45% allow anonymous enumeration:

$$150 \times 0.45 = 67.5 \approx 68$$

Among these, 60% are susceptible to SMB lateral movement:

$$67.5 \times 0.60 = 40.5 \approx 41$$

So, approximately 41 systems are initially exposed/susceptible to the SMB-based attack.

Second-stage lateral movement

Each compromised system attempts 3 systems, with success probability 0.3:

$$41 \times 3 \times 0.3 = 36.9$$

So approximately 37 additional systems may be compromised.

Estimated total:

$$41 + 37 = 78$$

Large-scale compromise?

70% of 150:

$$150 \times 0.70 = 105$$

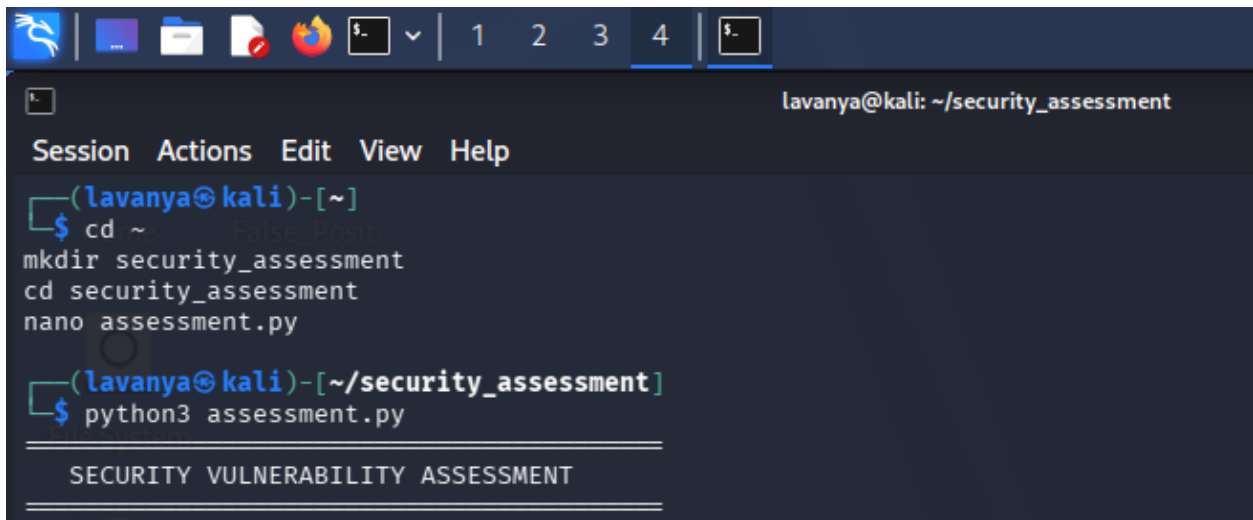
Estimated compromised systems = 78, which is below 105.

Therefore, based on this two-stage model, the attack does not exceed 70% of the network.

Priority mitigation

NULL sessions should be prioritized first, because they enable anonymous enumeration and provide the initial information/access that facilitates subsequent attacks. After that, weak SMB/shared-folder permissions should be hardened to reduce lateral movement.

Answer: Initial ≈ 41 systems, second-stage ≈ 37 additional systems, total ≈ 78 systems. This is about 52% of the network, so it does not cross the 70% threshold.



```
lavanya@kali: ~/security_assessment
Session Actions Edit View Help
(lavanya@kali)-[~]
$ cd ~
mkdir security_assessment
cd security_assessment
nano assessment.py
(lavanya@kali)-[~/security_assessment]
$ python3 assessment.py

=====
SECURITY VULNERABILITY ASSESSMENT
=====
```

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SECURITY VULNERABILITY ASSESSMENT

1. WINDOWS DOMAIN NETWORK

Initial susceptible systems : 40
Second-stage spread : 36
Estimated total : 77
Percentage compromised : 51.30%
Risk: Below 70% threshold
Priority: NULL sessions and weak SMB shares

2. RPC EXPOSURE

RPC exposed systems : 70
Vulnerable systems : 28
Expected compromises : 14
Expected detection alerts : 7
Assessment: Stealth attacker faces significant detection risk.

3. CLOUD LINUX SERVER

At least one vulnerable service : 64.30%
Full compromise probability : 32.15%
Assessment: File permissions alone are not sufficient.

4. NFS SECURITY ASSESSMENT

Unauthorized NFS access : 66.40%
Complete compromise : 46.48%
Restricted to one subnet: 14.00%
Assessment: Restricting NFS access significantly reduces risk.

5. SQL INJECTION ASSESSMENT

Records extracted : 20000
Database compromised : 40.00%
Sensitive records : 6000
Assessment: HIGH-SEVERITY BREACH
Firewall alone is NOT sufficient.

ASSESSMENT COMPLETE

2. RPC Exposure and Exploitation

Total systems = 120

RPC-exposed systems:

70

40% are vulnerable:

$$70 \times 0.40 = 28$$

Successful exploitation probability = 0.5:

$$28 \times 0.5 = 14$$

Expected compromised systems

14 systems

Expected detection alerts

Detection probability = 0.25.

Assuming each exploitation attempt is detected independently:

$$28 \times 0.25 = 7$$

Expected detection alerts = 7

We can also calculate detection among successful compromises:

$$14 \times 0.25 = 3.5$$

But the question asks for alerts during the exploitation process, so 7 alerts is the appropriate expected value when every vulnerable system is attempted.

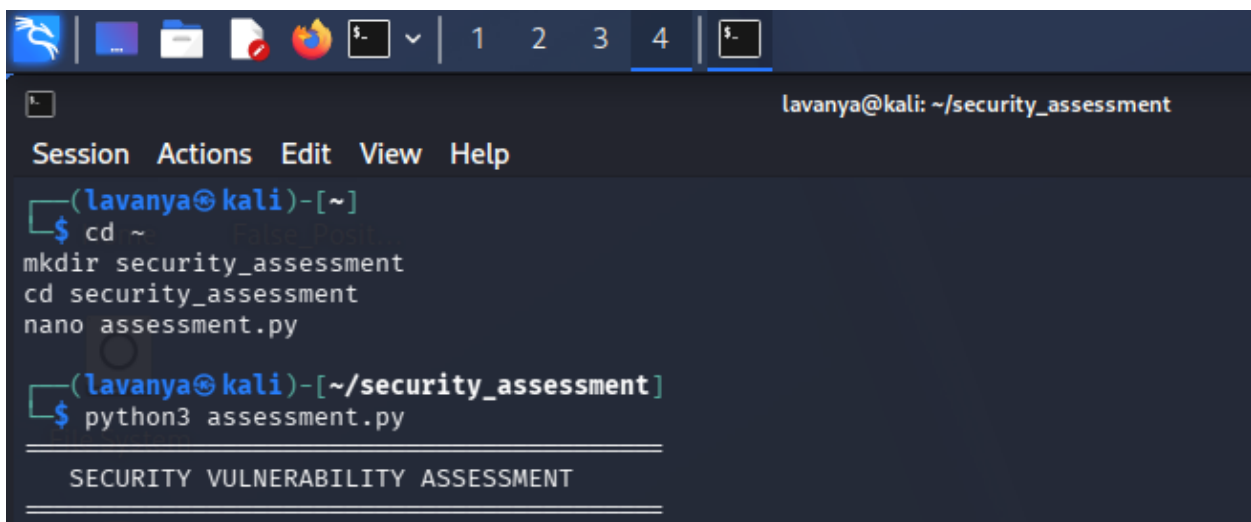
Security assessment

An attacker attempting all 28 vulnerable systems has:

- 14 expected successful compromises
- 7 expected detection alerts

A stealth-focused attacker would therefore have significant detection risk. They may prefer alternative strategies such as credential abuse, phishing, or less-monitored attack paths, rather than repeatedly exploiting exposed RPC services.

Answer: Expected compromise = 14 systems; expected alerts = 7. RPC exploitation presents a considerable detection risk for a stealth-oriented attacker.



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3. Cloud Linux Server

Four services have vulnerability probabilities:

- $P1=0.3$
- $P2=0.2$
- $P3=0.25$
- $P4=0.15$

We first calculate the probability that none are vulnerable.

$$P(\text{none})=(1-0.3)(1-0.2)(1-0.25)(1-0.15)=0.7\times0.8\times0.75\times0.85=0.357$$

Therefore:

$$P(\text{at least one})=1-0.357=0.643$$

So the probability that at least one service is vulnerable is:

64.3%

Full system compromise

Weak file permissions can be exploited with probability 0.5 after initial access.

$$0.643\times0.5=0.3215$$

Therefore:

32.15%

Assessment

A 64.3% probability of at least one vulnerable service is significant.

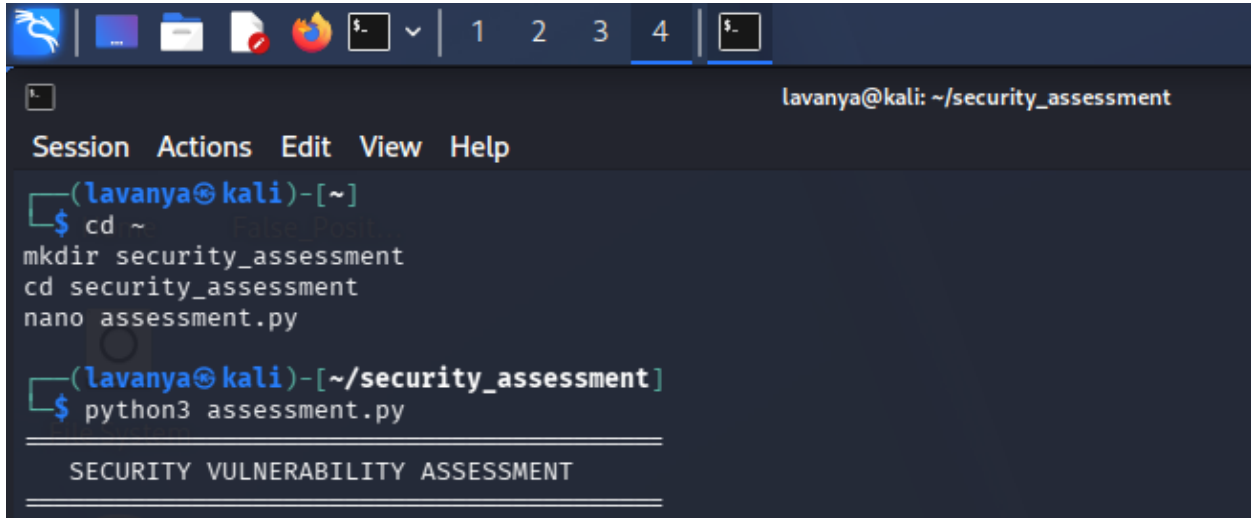
Strengthening file permissions alone reduces the probability of progressing from initial access to full compromise, but it does not remove the initial attack surface.

Therefore, the organization should:

1. Patch/harden the exposed services.

2. Strengthen file permissions.
3. Apply least privilege.
4. Restrict unnecessary internet exposure.

Answer: At least one service vulnerable = 64.3%. Overall full compromise probability = 32.15%. File-permission hardening helps significantly, but should not be the only mitigation.

A terminal window on a Kali Linux system. The window title is 'lavanya@kali: ~/security_assessment'. The terminal shows the following commands and output:

```
(lavanya@kali)-[~]  
$ cd ~  
mkdir security_assessment  
cd security_assessment  
nano assessment.py  
  
(lavanya@kali)-[~/security_assessment]  
$ python3 assessment.py  
  
=====
```

The output of the script is 'SECURITY VULNERABILITY ASSESSMENT', which is displayed between two sets of equals signs. The terminal interface includes a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The top of the window shows a taskbar with various application icons and a tab labeled '4'.

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4. NFS Unauthorized Access

Three subnets have unauthorized-access probabilities:

0.4,0.3,0.2

At least one subnet gains access

Calculate the probability that none succeeds:

$$(1-0.4)(1-0.3)(1-0.2) = 0.6 \times 0.7 \times 0.8 = 0.336$$

Therefore:

$$P(\text{at least one}) = 1 - 0.336 = 0.664$$

So:

66.4%

Complete system compromise

Privilege escalation probability = 0.7.

$$0.664 \times 0.7 = 0.4648$$

Therefore:

46.48%

Restricting NFS to one subnet

If access is restricted to the lowest-risk subnet, whose unauthorized-access probability is 0.2:

$$P(\text{complete compromise}) = 0.2 \times 0.7 = 0.14$$

So the risk becomes 14%, compared with 46.48%.

Risk reduction:

46.48–14=32.48 percentage points

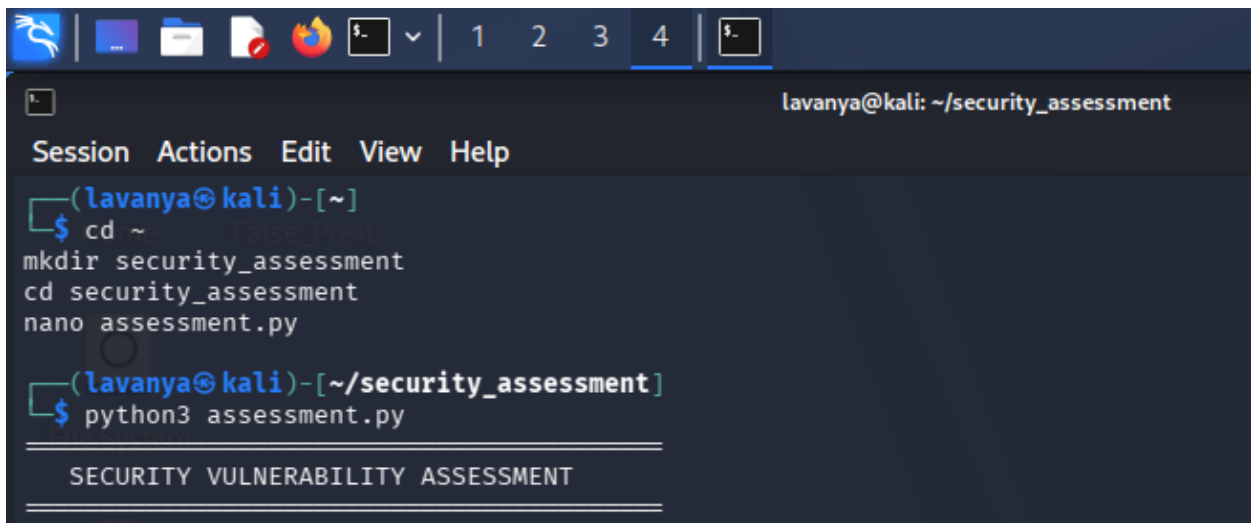
This is a substantial reduction.

Assessment

Restricting NFS access to a single trusted subnet is an effective mitigation, especially when combined with proper NFS export configuration, authentication and least privilege.

Answer: Unauthorized access = 66.4%; complete compromise = 46.48%.

Restricting access to the 20%-risk subnet reduces complete compromise probability to 14%.



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5. SQL Injection — E-commerce Database

Total records = 50,000

Extraction rate:

500 records/second

Firewall blocks attack after:

40 seconds

Records extracted

$$500 \times 40 = 20,000$$

Therefore:

20,000 records

Percentage of database compromised

$$50,000 \div 20,000 \times 100 = 40\%$$

So 40% of the database could be exposed.

Sensitive records

30% of the database is sensitive:

$$50,000 \times 0.30 = 15,000$$

Assuming the extracted records are representative of the database:

$$20,000 \times 0.30 = 6,000$$

Therefore approximately:

6,000 sensitive records

could be exposed.

Assessment

The firewall is useful because it limits the attack duration, but it is not sufficient protection.

In only 40 seconds, the attacker could extract:

- 20,000 total records
- 40% of the database
- approximately 6,000 sensitive records

This represents a high-severity breach.

The organization should additionally implement:

- Parameterized queries/prepared statements
- Input validation
- Web application firewall rules
- Database least privilege
- Continuous monitoring
- Rate limiting
- SQL injection testing

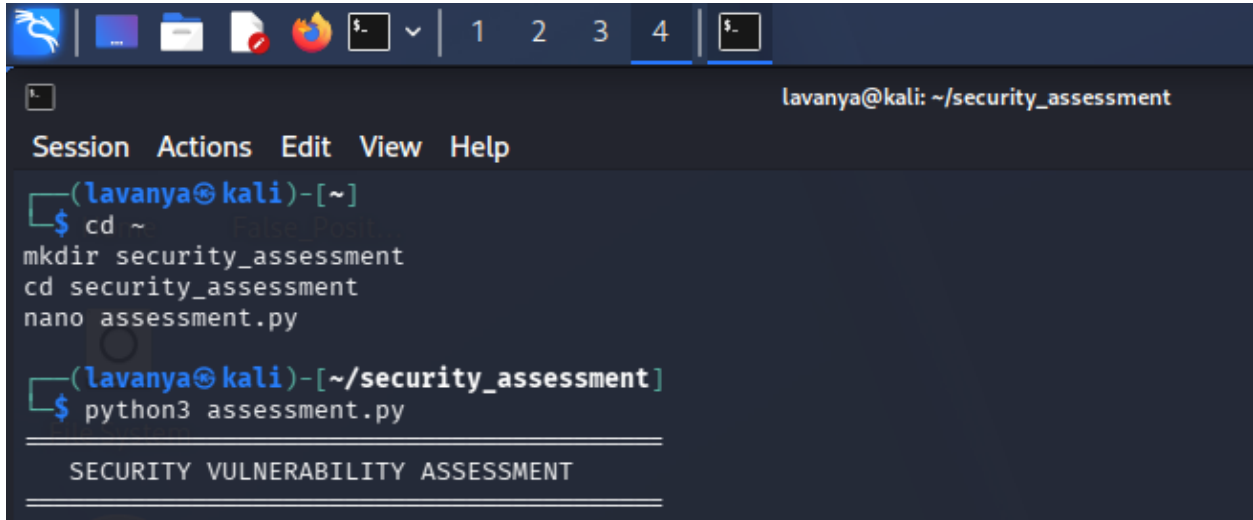
Final Answer

Question	Key Result	Security Assessment
1. Windows/SMB	~41 initial, ~37 additional, ~78 total	Below 70%, but serious lateral-movement risk
2. RPC	14 compromised, 7 alerts	Stealth attacker faces significant detection
3. Cloud Linux	64.3% vulnerable-service probability; 32.15% full compromise	Harden services and permissions
4. NFS	66.4% unauthorized access; 46.48% full compromise	Restricting NFS can greatly reduce risk

5. SQL Injection

20,000 records; 40%; ~6,000 sensitive

High severity; firewall alone insufficient



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